Hüseyin Felek

Embedded Software Engineer | Industrial Communication & Networking

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SUMMARY

Embedded software engineer with nearly 5 years of experience developing C/C++ software for industrial communication, networking and real-time embedded systems. Built L2/L3 networking features on a Linux-based switching and routing platform, including software that controls switch ASICs, and developed PROFINET and Modbus communication, device drivers and platform software for an embedded Linux automation controller deployed on 1000+ field devices. Experienced in real-time firmware on FreeRTOS, the full product life cycle from requirements and design to testing and release, code review, and Git-based workflows with GitHub and GitLab.

WORK HISTORY

Software Design Engineer · Andasis Elektronik 05.2025 – 02.2026

- Developed and stabilized core L2/L3 networking features on an OpenWrt-based switching and routing platform, including DHCP relay, IEEE 802.1X/RADIUS authentication, ARP inspection and IGMP, bringing each feature to production readiness.
- Developed software communicating with Marvell switch ASICs through the CPSS SDK to configure packet forwarding (VLAN, FDB, trapping, mirroring), and implemented TCAM-based ACL/PCL rules for packet filtering and classification.
- Extended SNMP/MIB support with new OIDs for actively developed protocols.
- Debugged reliability and throughput issues at packet level with Wireshark, and module-tested features during development.

Embedded Systems R&D Engineer · Eliar Elektronik 07.2021 – 05.2025

Embedded Linux and Industrial Communication

- Architected an ARMv8 embedded Linux automation controller platform with one hardware design and software-defined variants, from a PROFINET I/O coupler built on a third-party PROFINET device stack to a full CODESYS-based controller.
- Authored foundational C/C++ libraries and device drivers for I/O module communication with Modbus and PROFINET data exchange.
- Redesigned I/O scanning from full-cycle to change-based updates using per-device XML descriptors, deployed on 1000+ field devices replacing legacy controllers.
- Engineered time-critical software with multi-threading and concurrency primitives, and implemented power-backup logic against unexpected shutdowns.
- Implemented UDP and Bluetooth-based device discovery and inter-device communication, and built web-based interfaces for configuration, monitoring and OTA updates with versioning and image backup/restore.
- Developed APIs and an I/O simulator so application and GUI developers could test without hardware; containerized on-device services with Docker and managed systemd services for CODESYS runtime integration.

Real-Time Firmware and Wireless Communication

- Owned the firmware architecture for a family of industrial smart sensors (conductivity, pH/ORP, optical), developing real-time STM32 firmware under FreeRTOS with interrupt-driven event handling, task scheduling, priority assignment and latency optimization.
- Integrated MCU peripherals (ADC, DAC, DMA, SPI, I^2C, USART) and communication interfaces (TCP/IP, UDP, CAN, RS-485, RS-232, IO-Link) across the sensor product line.
- Customized the STM32 bootloader for secure over-the-air updates with SBSFU, with images received via ESP32 and relayed over Y-Modem for flash write and partition management.
- Led development of the shared ESP32 Bluetooth and Wi-Fi communication and configuration stack, and led certification testing for the company's first officially BLE-certified device.
- Defined specifications and designed the analog front-end for a new dual-channel pH/ORP product.

Testing and Product Life Cycle

- Tested each developed module with module-level tests and took part in the full product life cycle, from requirements and design through debugging, release and post-production quality assurance.
- Took part in code review, both reviewing teammates' code and having my own reviewed; used Git with GitHub and GitLab, and tracked work and documentation in Jira and Confluence.
- Built CODESYS-based and PC-based production test and calibration systems with automated test sequencing and data logging.

Intern · Siemens (Mobility) 07.2018 – 09.2018

CORE COMPETENCIES

Programming	C, C++, Python, Bash, Makefile
Embedded Linux & Networking	Embedded Linux, OpenWrt, systemd, Docker, L2/L3 switching and routing, VLAN, IEEE 802.1X/RADIUS, DHCP relay, IGMP, SNMP/MIB, Marvell CPSS, TCAM ACL
Real-Time & Firmware	FreeRTOS, real-time systems, bare-metal, STM32, ESP32, interrupt handling, DMA, bootloaders (SBSFU)
Industrial & Wireless Communication	PROFINET IO, Modbus, IO-Link, CAN, RS-485, RS-232, TCP/IP, UDP, MQTT, Bluetooth/BLE, Wi-Fi
Tools & Process	Git (GitHub, GitLab), code review, GDB, JTAG (ST-Link, J-Link), Wireshark, logic and protocol analyzers, module testing, familiarity with MISRA C, Jira, Confluence, Agile/Scrum

EDUCATION & LANGUAGES

Bachelor of Science, Electrical and Electronics Engineering · Boğaziçi University 2016 – 2021

Turkish Native | English Advanced | French Intermediate | Willing to travel